

Multi-Type Proximity Clustering and Heart Disease Analysis

Introduction

This project presents two Python-based clustering and proximity analysis systems developed for datasets with different structures and characteristics. The implementation focuses on distance metrics, clustering techniques, dimensionality reduction, and visualization.

The project was developed using: Python, pandas, scikit-learn, matplotlib, seaborn, gower, openpyxl

Software 1: Multi-Type Proximity Clustering

The first software tool analyzes small datasets containing:

- Nominal data
- Binary data
- Ordinal data
- Cardinal numerical data
- Mixed-type data

Different proximity metrics were applied depending on the data type:

- Hamming Distance
- Jaccard Distance
- Euclidean Distance
- Gower Distance

K-Means clustering was performed for all datasets, while PCA was used for visualization. The software automatically exports:

- Distance matrices
- Cluster assignments
- PCA plots
- Excel reports with embedded visualizations

The results showed that:

- Nominal and Binary datasets produced clearer cluster separation.
- Gower distance performed effectively for mixed-type data.

Software 2: Heart Disease Clustering Analysis

The second software tool was applied to a larger heart disease dataset containing medical attributes such as:

- age
- cholesterol
- blood pressure
- heart rate
- ECG results
- exercise-induced angina

The following distance metrics were compared:

- Euclidean
- Cityblock (Manhattan)
- Cosine
- Correlation

The workflow included:

- Data normalization
- Distance matrix computation
- t-SNE dimensionality reduction
- K-Means clustering
- Silhouette Score evaluation

The Correlation and Cosine distances produced the best clustering performance for the dataset.

Conclusion

The experiments demonstrated that clustering performance strongly depends on the selected proximity metric and the structure of the dataset.

Gower distance proved effective for heterogeneous mixed-type datasets, while Correlation and Cosine distances produced better cluster separation in the heart disease dataset.

Dimensionality reduction techniques such as PCA and t-SNE significantly improved the interpretability of clustering structures and visual analysis.

The project highlights the importance of selecting suitable similarity measures for different machine learning and data analysis scenarios.

References

- Scikit-learn Documentation
- UCI Heart Disease Dataset
- t-SNE Research Paper (van der Maaten & Hinton, 2008)
- Gower Distance Documentation

